A positive answer to Dales Question (II) for $c_0 \oplus \ell_\infty$

Smoke-test packet for `fa_banach_001`

2026-06-14

Status

This packet claims a full solution of the specific subquestion in Dales–Kania–Kochanek–Koszmider–Laustsen [1]: for

$$E = c_0 \oplus \ell_\infty,$$

every finitely-generated maximal left ideal of $\mathscr{B}(E)$ is fixed. It does not address the remaining $C(K)$ cases in [1], such as $C[0,1]$ or $C[0,\alpha]$ for $\alpha \geq \omega^\omega$.

Source Question

In [1, Question (II)], Dales asks whether every finitely-generated maximal left ideal of $\mathscr{B}(E)$ is fixed. The same paper states on pages 23–24 that the answer was not known for $E = c_0 \oplus \ell_\infty$.

> Then E^* is isomorphic to $\big(\bigoplus_{n\in\mathbb{N}} E_n^*\big)_{\ell_\infty}$, and so the conclusion is that each finitely-generated, maximal left ideal of $\mathscr{B}\big(\big(\bigoplus E_n^*\big)_{\ell_\infty}\big)$ is fixed.
>
> The conditions imposed on the Banach space E in Theorems 5.8 and 5.11 are clearly preserved under the formation of finite direct sums. In contrast, this need not be the case for the condition of Corollary 5.3. For instance, c_0 and ℓ_∞ both satisfy this condition by Example 5.4(ii)–(iii), whereas their direct sum $c_0 \oplus \ell_\infty$ does not. We shall explore this situation in greater depth in Section 7. Notably, as a particular instance of Theorem 7.3, we shall see that the main conclusion of Corollary 5.3 fails for $E = c_0 \oplus \ell_\infty$ because $\mathscr{F}(c_0 \oplus \ell_\infty)$ *is* contained in a proper, closed, singly-generated left ideal of $\mathscr{B}(c_0 \oplus \ell_\infty)$.
>
> We do not know the answer to Question (II) for $E = c_0 \oplus \ell_\infty$, but the following result answers this question positively for another direct sum arising

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naturally from Example 5.4, with the ideal $\mathscr{S}(E)$ of strictly singular operators taking the role that was played by $\mathscr{F}(E)$ in Corollary 5.3 and $\mathscr{K}(E)$ in Theorem 5.8.

Proposition 5.13. *Let $E = c_0(\mathbb{I}) \oplus H$, where $\mathbb{I}$ is a non-empty set and H is a Hilbert space. Then $\mathscr{B}(E)$ is the only finitely-generated left ideal of $\mathscr{B}(E)$*

Definitions and Source Inputs

A maximal left ideal of $\mathscr{B}(E)$ is fixed if it has the form

$$\mathscr{M}\mathscr{L}_x = \{T \in \mathscr{B}(E) : Tx = 0\}$$

for some non-zero $x \in E$. We use the following results from [1].

- The dichotomy theorem says that a non-fixed maximal left ideal contains $\mathscr{F}(E)$.
- The strong dichotomy, Corollary 4.1, says that a non-fixed maximal left ideal contains the ideal $\mathscr{E}(E)$ of inessential operators.
- Corollary 4.7 says that, for a finite generating family $\Gamma \subset \mathscr{B}(E)$, containment $\mathscr{F}(E) \subseteq \mathscr{L}_\Gamma$ is equivalent to the associated map

  $$\Psi_\Gamma : E \to E^{|\Gamma|}, \qquad \Psi_\Gamma x = (Tx)_{T\in\Gamma},$$

  being bounded below.
- Lemma 7.1 says that every operator $\ell_\infty \to X$ is strictly singular whenever X contains no copy of ℓ_∞.
- Proposition 7.8 says that, for $E = c_0 \oplus \ell_\infty$, the ideal

  $$\mathscr{K}_1 = \left\{ \begin{pmatrix} A & B \\ C & D \end{pmatrix} \in \mathscr{B}(c_0 \oplus \ell_\infty) : A \in \mathscr{K}(c_0) \right\}$$

  is not contained in any finitely-generated maximal left ideal of $\mathscr{B}(E)$.

Proof Intuition

A non-fixed finitely-generated maximal left ideal L would contain $\mathscr{F}(E)$, so the finite generator map is bounded below. Since $c_0 \oplus \ell_\infty$ is isomorphic to each of its finite powers, this places a bounded-below operator inside L. The strong dichotomy also places all inessential operators inside L. For $c_0 \oplus \ell_\infty$, the off-diagonal blocks are inessential, and the ℓ_∞-to-c_0 block of a bounded-below operator is strictly singular. Therefore the lower right block is upper semi-Fredholm, and injectivity of ℓ_∞ lets one manufacture the lower-right identity projection inside L. Together with the inessential blocks this forces $\mathscr{K}_1 \subseteq L$, contradicting Proposition 7.8 of the source paper.

Lemma 1. *Let $E = c_0 \oplus \ell_\infty$. If L is a non-fixed maximal left ideal of $\mathscr{B}(E)$ and L contains a bounded-below operator, then $\mathscr{K}_1 \subseteq L$.*

Proof. Write operators on E in block form relative to $E = c_0 \oplus \ell_\infty$. Since L is non-fixed, the strong dichotomy [1, Corollary 4.1] gives $\mathscr{E}(E) \subseteq L$.

We first record the block consequences. Compact operators on c_0 are inessential. Every operator $\ell_\infty \to c_0$ is strictly singular by [1, Lemma 7.1], because c_0 contains no copy of ℓ_∞; hence such operators are inessential. By the standard symmetry property for inessential operators, used in [1, proof of Theorem 7.7] via Gonzalez [2], every operator $c_0 \to \ell_\infty$ is also inessential. Finally, weakly compact operators on ℓ_∞ are inessential. Thus L contains every block matrix

$$\begin{pmatrix} A & B \\ C & D \end{pmatrix} \quad \text{with} \quad A \in \mathscr{K}(c_0),\ B \in \mathscr{B}(\ell_\infty, c_0),\ C \in \mathscr{B}(c_0, \ell_\infty),\ D \in \mathscr{W}(\ell_\infty).$$

Let

$$R = \begin{pmatrix} R_{11} & R_{12} \\ R_{21} & R_{22} \end{pmatrix} \in L$$

be bounded below. Restricting R to the ℓ_∞-summand shows that the column map

$$y \mapsto (R_{12}y, R_{22}y), \qquad \ell_\infty \to c_0 \oplus \ell_\infty,$$

is bounded below, hence upper semi-Fredholm. The operator R_{12} is strictly singular, so stability of upper semi-Fredholm operators under strictly singular perturbations gives that R_{22} is upper semi-Fredholm.

Choose a finite-rank projection $Q \in \mathscr{B}(\ell_\infty)$ onto $\ker R_{22}$. The restriction of R_{22} to $\ker Q$ is an isomorphism onto its closed range. Since ℓ_∞ is injective, the inverse on $R_{22}(\ker Q)$ extends to an operator $S \in \mathscr{B}(\ell_\infty)$, and so

$$SR_{22} = I_{\ell_\infty} - Q.$$

As $R \in L$, we have

$$\begin{pmatrix} 0 & 0 \\ 0 & S \end{pmatrix} R = \begin{pmatrix} 0 & 0 \\ SR_{21} & I_{\ell_\infty} - Q \end{pmatrix} \in L.$$

The off-diagonal block $SR_{21} : c_0 \to \ell_\infty$ is inessential, and Q is finite-rank, hence weakly compact. Both corresponding block operators belong to L. Therefore

$$P_2 := \begin{pmatrix} 0 & 0 \\ 0 & I_{\ell_\infty} \end{pmatrix} \in L.$$

Left-multiplying P_2 by arbitrary lower-right block matrices gives every operator of the form

$$\begin{pmatrix} 0 & 0 \\ 0 & D \end{pmatrix}, \qquad D \in \mathscr{B}(\ell_\infty).$$

Adding the inessential upper-left compact and off-diagonal blocks recorded above yields $\mathscr{K}_1 \subseteq L$. □

Theorem 1. *Every finitely-generated maximal left ideal of $\mathscr{B}(c_0 \oplus \ell_\infty)$ is fixed.*

Proof. Let $E = c_0 \oplus \ell_\infty$, and suppose toward a contradiction that L is a non-fixed finitely-generated maximal left ideal of $\mathscr{B}(E)$. Choose a finite generating set Γ with $L = \mathscr{L}_\Gamma$. By the dichotomy theorem, $\mathscr{F}(E) \subseteq L$. Corollary 4.7 of [1] implies that $\Psi_\Gamma : E \to E^{|\Gamma|}$ is bounded below.

For every $n \in \mathbb{N}$, $c_0^n \cong c_0$ and $\ell_\infty^n \cong \ell_\infty$, hence $E^n \cong E$. Taking an isomorphism $A : E^{|\Gamma|} \to E$, the operator

$$R = A\Psi_\Gamma$$

is bounded below and belongs to L. Lemma 1 gives $\mathscr{K}_1 \subseteq L$. This contradicts [1, Proposition 7.8], which states that $\mathscr{K}_1$ is not contained in any finitely-generated maximal left ideal of $\mathscr{B}(c_0 \oplus \ell_\infty)$. Thus no such non-fixed L exists. □

Verification and Novelty Check

No computation is used. The proof is a formal combination of source-paper theorems plus Lemma 1. The local run indexes

`registry_index.tsv`, `solutions/index.tsv`
`attempts/index.tsv`, `proof_gaps/index.tsv`

were searched for arXiv `1208.4762`, DalesQ1, $c_0 \oplus \ell_\infty$, and maximal-left-ideal keywords; no duplicate packet was found. Web searches on 2026-06-14 for exact phrases involving DalesQ1, $c_0 \oplus \ell_\infty$, $C[0,1]$, and finitely-generated maximal left ideals found the source paper but no later exact answer.

Novelty confidence is moderate. The packet relies heavily on [1], especially Proposition 7.8, and should be reviewed as a short corollary-style completion of the stated $c_0 \oplus \ell_\infty$ subquestion. It does not settle the broader $C(K)$ cases.

References

[1] H. G. Dales, T. Kania, T. Kochanek, P. Koszmider, and N. J. Laustsen, *Maximal left ideals of the Banach algebra of bounded operators on a Banach space*, arXiv:1208.4762v3.

[2] M. Gonzalez, *On essentially incomparable Banach spaces*, Math. Z. **215** (1994), 621–629.

Human Review: Dales Question (II) for $c_0 \oplus \ell_\infty$

Original automatic proof: `solution_packet.pdf`
Checked date: 15 June 2026
Status: Verified.
Verdict: The proof appears essentially correct. It gives a positive answer to the stated subquestion for

$$E = c_0 \oplus \ell_\infty,$$

namely that every finitely-generated maximal left ideal of $\mathcal{B}(E)$ is fixed.

Human Comments

The proof has a clear structure. The main new step is the following implication: if L is a non-fixed maximal left ideal of $\mathcal{B}(c_0 \oplus \ell_\infty)$ and L contains a bounded-below operator, then

$$\mathcal{K}_1 \subseteq L,$$

where

$$\mathcal{K}_1 = \left\{ \begin{pmatrix} A & B \\ C & D \end{pmatrix} \in \mathcal{B}(c_0 \oplus \ell_\infty) : A \in \mathcal{K}(c_0) \right\}.$$

This is then combined with Proposition 7.8 of the source paper, which states that $\mathcal{K}_1$ is not contained in any finitely-generated maximal left ideal of $\mathcal{B}(c_0 \oplus \ell_\infty)$.

The block argument looks sound. Since L is non-fixed, the strong dichotomy in the source paper gives

$$\mathcal{E}(E) \subseteq L,$$

where $\mathcal{E}(E)$ is the ideal of inessential operators. The required block inclusions are justified as follows. Compact operators on c_0 are inessential. Every operator $\ell_\infty \to c_0$ is strictly singular, because c_0 contains no copy of ℓ_∞, and hence is inessential. The converse off-diagonal direction $c_0 \to \ell_\infty$ is also inessential by the standard symmetry property for inessential operators, exactly as used in the proof of Theorem 7.7 of the source paper via Gonzalez. Finally, weakly compact operators on ℓ_∞ are inessential.

Now let

$$R = \begin{pmatrix} R_{11} & R_{12} \\ R_{21} & R_{22} \end{pmatrix} \in L$$

be bounded below. Restricting R to the ℓ_∞-summand gives the column operator

$$y \mapsto (R_{12}y, R_{22}y), \qquad \ell_\infty \to c_0 \oplus \ell_\infty,$$

which is bounded below and hence upper semi-Fredholm. Since $R_{12} : \ell_\infty \to c_0$ is strictly singular, stability of upper semi-Fredholm operators under strictly singular perturbations implies that R_{22} is upper semi-Fredholm.

Choosing a finite-rank projection Q onto $\ker R_{22}$, the restriction of R_{22} to $\ker Q$ is an isomorphism onto its closed range. Since ℓ_∞ is injective, the inverse on this range extends to an operator $S \in \mathcal{B}(\ell_\infty)$, giving

$$SR_{22} = I_{\ell_\infty} - Q.$$

Multiplying R on the left by the lower-right block operator with entry S then produces

$$\begin{pmatrix} 0 & 0 \\ SR_{21} & I_{\ell_\infty} - Q \end{pmatrix} \in L.$$

The lower-left block SR_{21} is inessential, and Q is finite-rank. Since these corresponding block operators already belong to L, it follows that

$$\begin{pmatrix} 0 & 0 \\ 0 & I_{\ell_\infty} \end{pmatrix} \in L.$$

This forces all lower-right blocks into L, and together with the inessential upper-left compact and off-diagonal blocks gives $\mathcal{K}_1 \subseteq L$.

It remains to explain why a non-fixed finitely-generated maximal left ideal contains such a bounded-below operator. If

$$L = \mathscr{L}_\Gamma, \qquad \Gamma = \{T_1, \ldots, T_n\},$$

then the dichotomy theorem gives $\mathcal{F}(E) \subseteq L$. Corollary 4.7 of the source paper then implies that

$$\Psi_\Gamma : E \to E^n, \qquad \Psi_\Gamma x = (T_1 x, \ldots, T_n x),$$

is bounded below. Since $E^n \cong E$, choose an isomorphism $A : E^n \to E$. Then

$$R = A\Psi_\Gamma$$

is bounded below. Moreover $R \in L$, because writing A coordinatewise as

$$A(y_1, \ldots, y_n) = A_1 y_1 + \cdots + A_n y_n$$

gives

$$A\Psi_\Gamma = A_1 T_1 + \cdots + A_n T_n \in \mathscr{L}_\Gamma = L.$$

Thus a non-fixed finitely-generated maximal left ideal would contain a bounded-below operator, hence would contain $\mathcal{K}_1$, contradicting Proposition 7.8. The conclusion follows.

Review Outcome

Accepted as an essentially correct proof of the $c_0 \oplus \ell_\infty$ subcase. The result should be viewed as a corollary-style completion of this specific open subquestion using the machinery already developed in the source paper, rather than as a solution to the broader remaining $C(K)$ cases.